



Step-by-Step: Formulas for success

This reading outlines the steps the instructor performs in the next video, [Formulas for success](#). In the video, the instructor explains the basics of using spreadsheet formulas for calculations.

Keep this step-by-step guide open as you watch the video. It can serve as a helpful reference if you need additional context or clarification while following the video steps. This is not a graded activity, but you can complete these steps to practice the skills demonstrated in the video.

What you'll need

If you'd like to follow along with the first example in this video, choose a spreadsheet tool and open a blank sheet.

If you would like to access the other spreadsheet the instructor uses in this video, click the link to the dataset to create a copy. If you don't have a Google account, download the data directly from the attachments below.

Link to sales data: [Monthly sales](#)

OR

[Monthly Sales](#)

[XLSX File](#)

Example 1: Create a formula

Formulas form the groundwork for more complex spreadsheet tasks. Here's a simple exercise:

1. Open a new spreadsheet.
2. Select cell A1.
3. Enter `=2-2` and then press Enter. The cell displays the result of 0.
4. Select cell A2.
5. Enter `=31982-17795` and press Enter. The cell displays the result 14187.

Note: The equal sign (=) signifies that you're beginning a formula.

Example 2: Use cell references in a formula

Cell references make your spreadsheet flexible and responsive to data changes. To implement this:

1. Open the [Monthly sales](#) spreadsheet.
2. To find the total sales for April through July of 2017, select cell F2.
3. Enter the formula `=B2+C2+D2+E2` and then press Enter. You now have the total sales for this timeframe.
4. But what if the data in one of the cells was incorrect? Select cell D2 and enter 47002 to correct the entry. Press Enter.
5. Notice that your spreadsheet automatically recalculates the sum in cell F2.

Example 3: Copy a formula

Copying and pasting formulas saves time and helps ensure consistency across your calculations. To do this:

1. In the Monthly sales spreadsheet, select cell F2.
2. From the Edit menu, choose Copy. You can also use the Windows keyboard shortcut Ctrl+C or Mac keyboard shortcut Command+C to copy the formula.
3. Select cell F3 and from the Edit menu, choose Paste. Alternatively, press Ctrl+V (Windows) or Command+V (Mac) to paste the formula into cell F3.

Note: After pasting into cell F3, the formula in that cell will be `=B3+C3+D3+E3`.

Example 4: Calculate the average sales

Use formulas for different calculations, such as finding an average:

1. In the Monthly sales spreadsheet, select cell G1.
2. Enter Average Sales in cell G1 and press Enter.
3. Select cell G1 again. In the toolbar, select Bold to bold the text.

Note: Naming

columns in spreadsheets enhances clarity by indicating the purpose of the numbers.

4. Select cell G2.

Enter `=(B2+C2+D2+E2) / 4` and press Enter to calculate the average sales over this timeframe.

5. Note: This formula calculates the average of sales for the month, including cases in which there are no sales (blank cells), which are treated as zeroes in the calculation. If the business had zero sales for a month, the blank cell is still included in the calculation to maintain accuracy.
6. Copy and paste the formula from cell G2 into cells G3 and G4.

Example 5: Calculate the percent change in sales

Use a different formula to calculate percent change:

1. In the Monthly sales spreadsheet, select cell H1.
2. In cell H1, enter June to July Change. Bold this text.
3. In cell H2, enter `= (E2-D2) / D2` to calculate the percentage change in sales.
4. To format the value as a percentage, on the toolbar, select the % button. You now find that the percent change in sales between June and July is 247.5%.
5. Copy this formula to cell H3. Notice that the spreadsheet copies both the formula and the percentage formatting.

Example 6: Correct a formula error

Correcting formula errors ensures your data analysis remains accurate and trustworthy.

Here's how to troubleshoot one common mistake:

1. In the Monthly sales spreadsheet, copy the formula from cell H2 to cell H4 and press Enter.
2. Notice the error displayed in cell H4. This error occurs because the formula is trying to divide by a value of zero. The reason for this error is that cell D4 is blank, and in this context, the spreadsheet interprets it as having a value of zero.
3. To resolve the error, type 75866 in cell D4 and press Enter. Notice that the error disappears, and cell H4 now displays 121.16%.